

Microbiological quality of retail cheeses made from raw, thermized or pasteurized milk in the UK

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Abstract

Two studies of retail fresh, ripened and semi-hard cheeses made from raw, thermized or pasteurized milk were undertaken in the UK during 2004 and 2005 to determine the microbiological quality of these products. Using microbiological criteria in European Commission Recommendations 2004/24/EC and 2005/175/EC, 2% of both raw, thermized (37/1819 samples) and pasteurized (51/2618 samples) milk cheeses were of unsatisfactory quality. Raw or thermized milk cheeses were of unsatisfactory quality due to levels of *Staphylococcus aureus* at $\geq 10^4$ cfu g⁻¹, *Escherichia coli* at $\geq 10^5$ cfu g⁻¹, and/or *Listeria monocytogenes* at $\geq 10^2$ cfu g⁻¹, whereas pasteurized milk cheeses were of unsatisfactory quality due to *S. aureus* at $\geq 10^3$ cfu g⁻¹ and/or *E. coli* at $\geq 10^3$ cfu g⁻¹. *Salmonella* was not detected in any samples. Cheeses were of unsatisfactory quality more frequently when sampled from premises rated as having little or no confidence in management and control systems, and stored/displayed at above 8 °C. Raw or thermized milk cheeses were also more likely to be of unsatisfactory quality when they were unripened types, and pasteurized milk cheeses when they were: semi-hard types; from specialist cheese shops or delicatessens; cut to order. These results emphasize the need for applying and maintaining good hygiene practices throughout the food chain to prevent contamination and/or bacterial growth. Labelling of cheeses with clear information on whether the cheese was prepared from raw milk also requires improvement.

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1. Introduction

Cheesemaking is a major industry worldwide, and much is still practised on a relatively small scale which accounts for the rich diversity of cheeses available (Fox and McSweeney, 2004). Classification of cheeses is made difficult by this diversity but the most widely accepted approach is one based on moisture content, with further subdivision depending on milk type and the role of microorganisms in cheese ripening. The attribute of ‘softness’ or ‘hardness’ is therefore directly related to the moisture content of the cheese, higher moisture cheeses being softer than low moisture cheeses. Cheese consists

primarily of milk fat and coagulated proteins and preservation is largely achieved by controlling the pH and water activity. Other intrinsic parameters that may affect the growth and survival of microorganisms in cheeses are redox potential and the presence of antimicrobial compounds produced by starter and non-starter organisms (Beresford and Williams, 2004). These properties of cheese, together with the length of maturation of the finished product and the fact that they are normally stored at a controlled temperature, constitute a ‘hurdle’ system of preservation that act as control steps to inhibit the growth of pathogenic bacteria.

Whilst the efficient pasteurization of milk (high temperature/short time; 72 °C for at least 15 s or low temperature/long time; 63 °C for at least 30 min) should eliminate the risk from viable vegetative pathogenic

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